

Qatar-1b

R-0592

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_241007 · created 2026-07-24T23:31:05+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460590.7376 +/- 0.0028 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.14 +/- 0.032
Transit depth (Rp/Rs)^2	1.96 +/- 0.89 [%]
Orbital Inclination (inc)	82.42 +/- 2.06
Airmass coefficient 1 (a1)	0.997 +/- 0.013
Airmass coefficient 2 (a2)	0.004 +/- 0.014
Scatter in the residuals of the lightcurve fit is	1.28 %
Best Comparison Star	#2 - [480, 359]
Optimal Aperture	3.2
Optimal Annulus	13.68
Transit Duration (day)	0.052 +/- 0.021

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.14 $\pm$ 0.032 vs 0.1463 (deviation 0.2 σ)
Duration fit vs published	0.052 d vs 0.0692 d (deviation 0.82 σ)
Mid-time O-C	-0.7 min
Depth SNR	4.4
Residual scatter	1.28 %

Light curve

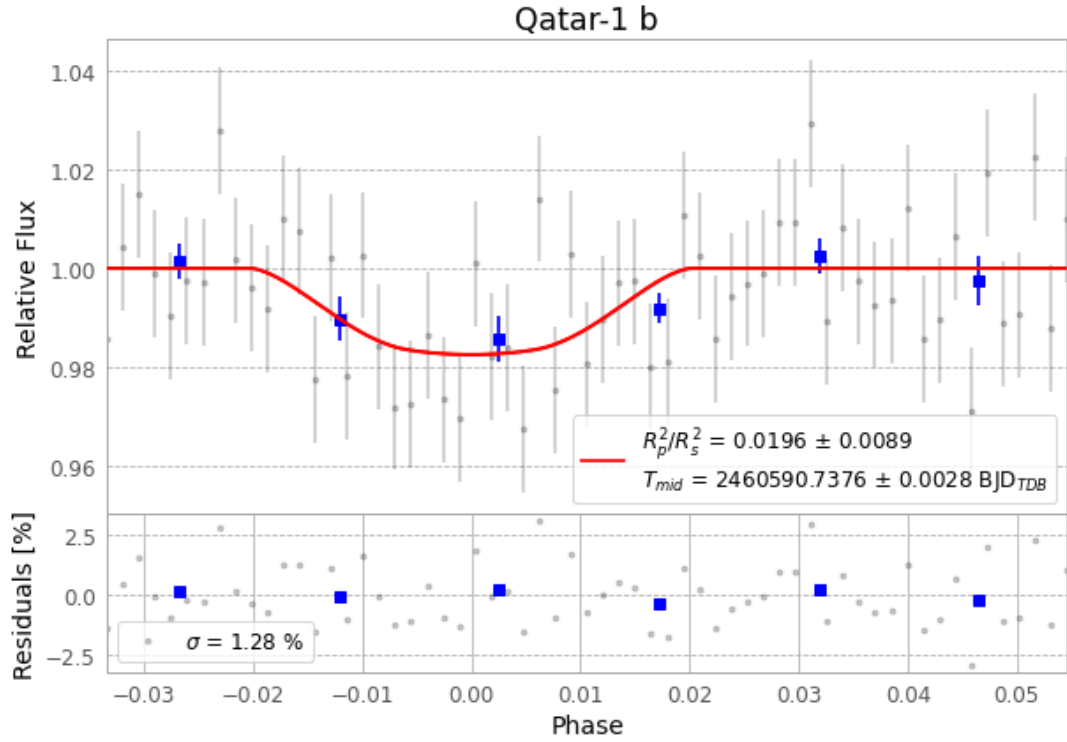


Figure 1 — FinalLightCurve_Qatar-1 b_2024-10-06.png (SHA-256 ceb097436e2bccd6...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [364, 280], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 faf7bbcf55386249...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

61 FITS frames (40 MB), Qatar-1241007043015.FITS → Qatar-1241007073016.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-241007.log (f8cbb55ce1ee...), FinalLightCurve_Qatar-1 b_2024-10-06.png (ceb097436e2b...), FinalLightCurve_Qatar-1 b_2024-10-06.csv (df5360c1a71c...)

Compute resources

Wall time 105.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-24 23:31Z.